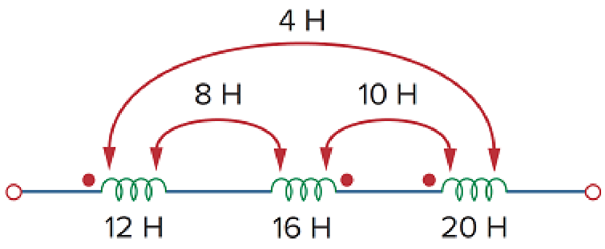
Problem

For the three coupled coils in Fig. 13.72, calculate the total inductance.

Figure 13.72



Step-by-step solution

Step 1 of 5

Figure 1

Step 2 of 5

Calculate the value of the equivalent inductance L_{eq1} for coil 1.

Write the expression for equivalent inductance L_{eq1} for coil 1.

$$\begin{aligned} L_{eq1} &= L_1 - M_{12} + M_{13} \\ &= 12 - 8 + 4 \\ &= 8 \end{aligned}$$

Therefore, the value of the equivalent inductance L_{eq1} is 8 H.

Step 3 of 5

Calculate the value of the equivalent inductance L_{eq2} for coil 2.

Write the expression for equivalent inductance L_{eq1} for coil 2.

$$\begin{aligned} L_{eq2} &= L_2 - M_{21} - M_{23} \\ &= 16 - 8 - 10 \\ &= -2\text{ H} \end{aligned}$$

Therefore, the value of the equivalent inductance L_{eq1} is -2 H.

Step 4 of 5

Calculate the value of the equivalent inductance L_{eq3} for coil 3.

Write the expression for equivalent inductance L_{eq3} for coil 3.

$$\begin{aligned} L_{eq3} &= L_3 - M_{23} + M_{13} \\ &= 20 - 10 + 4 \\ &= 14 \end{aligned}$$

Therefore, the value of the equivalent inductance L_{eq3} is $14\ \mathrm{H}$.

Step 5 of 5

Calculate the value of the total equivalent inductance L_{eq}.

Write the expression for total equivalent inductance L_{eq}.

$$\begin{aligned} L_{eq} &= L_{eq1} + L_{eq2} + L_{eq3} \\ &= 8 - 2 + 14 \\ &= 20 \end{aligned}$$

Therefore, the value of total equivalent inductance L_{eq} is $\boxed{20\ \mathrm{H}}$.